

Rimányi Positivity for Morin Thom Polynomials

Audited results, exact obstructions, and the remaining all-rank gap

Research summary of the multi-agent proof search

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Abstract

For the Morin algebra $A_d = \mathbb{C}[t]/(t^{d+1})$, Rimányi’s conjecture predicts nonnegative coefficients of $\mathrm{Tp}_{A_d}^\ell$ in the ordinary Chern-monomial basis for every $d \geq 1$ and $\ell \geq 0$. This note records the results of an extended, adversarially audited proof search based on the Bérczi–Szenes iterated-residue formula.

The search did *not* produce a proof for arbitrary d . It did produce several rigorous general results: an exact orbit-packet description of every Chern coefficient and of the universal residue kernel; an equivalent 0-Hecke folding and h -positivity formulation; a curvilinear-incidence formula; a multiplicity-one hyperplane identity; and positivity, for every d, ℓ , of all coefficients whose largest Chern index is at most $\ell + 2$. It also gives a complete all-level proof in ranks $d \leq 5$. In rank five the original chamber series has a negative Laurent coefficient, but the correction GBC is proved to lie in the universal packet kernel by two chamber-safe contour swaps, and the kernel-equivalent representative GDJ has an explicit coefficientwise-positive Stanley decomposition.

The note also records exact counterexamples to the principal tempting all-rank mechanisms. These counterexamples sharply distinguish Chern positivity from Laurent positivity, Schur positivity, effectivity, nefness, and positivity of individual degeneration components.

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1 Executive conclusion

Put $L = \ell + 1$. The target conjecture is

$$\mathrm{Tp}_{A_d}^{L-1} \in \mathbb{Z}_{\geq 0}[c_1, c_2, \dots] \quad (d, L \geq 1).$$

The outcome of the proof search is summarized as follows.

Theorem 1.1 (Audited positive results). *The following statements were established.*

- (i) *For every $L \geq 1$, the Chern-monomial coefficients of $\mathrm{Tp}_{A_d}^{L-1}$ are nonnegative for $1 \leq d \leq 5$.*
- (ii) *For every $d, L \geq 1$, if $\lambda \vdash dL$ and $\lambda_1 \leq L + 1$, then*

$$[c\lambda] \mathrm{Tp}_{A_d}^{L-1} \geq 0.$$

- (iii) *Every Chern coefficient is a distinct-permutation orbit sum of coefficients of one fixed chamber Laurent series. Consequently the all-level universal residue kernel is exactly the kernel of the orbit-sum map.*

- (iv) *In rank five,*

$$\mathcal{Q}_5 = G(DJ + BC) = GDJ + GBC,$$

the summand GBC is universally residue-null for every L , and the fixed-chamber expansion associated with GDJ is coefficientwise nonnegative.

Unresolved point. No construction or identity was found that proves the full nonnegativity statement for arbitrary d . In particular, the results below are not a reduction of the full conjecture to a proved lemma. The all- d gap remains the positivity of the globally symmetrized chamber packets outside the sector $\lambda_1 \leq L + 1$.

The strongest surviving approaches at the stopping point were:

1. coefficient-array folding by the 0-Hecke algebra;
2. global circuit straightening modulo the full symmetric balancing ideal;
3. a history-sensitive algebra for nested curvilinear incidences;
4. a description of the *saturated* Morin orbit multidegree by nonlocal associativity circuits.

Each is described below together with its exact current obstruction.

2 Residue conventions and the chamber series

Let

$$\Delta_d(z) = \prod_{1 \leq i < j \leq d} (z_i - z_j)$$

and

$$\mathcal{D}_d(z) = \prod_{l=1}^d \prod_{m=1}^{l-1} \prod_{r=1}^{\min(m, l-m)} (z_m + z_r - z_l).$$

The Bérczi–Szenes formula is

$$\mathrm{Tp}_{A_d}^\ell = (-1)^d \mathrm{IRes}_C \frac{\Delta_d(z) \mathcal{Q}_d(z)}{\mathcal{D}_d(z)} \prod_{i=1}^d c(1/z_i) z_i^\ell dz_i, \quad (1)$$

where

$$C : \quad |z_1| \ll |z_2| \ll \cdots \ll |z_d|, \quad c(u) = \sum_{j \geq 0} c_j u^j,$$

and throughout

$$c_0 = 1, \quad c_j = 0 \quad (j < 0).$$

Write $\mathcal{E}_C$ for expansion in this chamber. Our residue convention is

$$\mathrm{Res}_{z=\infty} f(z) dz = -[z^{-1}] f(z).$$

Thus, for a chamber-expanded d -form,

$$\mathrm{IRes}_C f(z) dz_1 \cdots dz_d = (-1)^d [z_1^{-1} \cdots z_d^{-1}] \mathcal{E}_C f.$$

The d residue signs cancel the displayed factor $(-1)^d$ in (1). This cancellation is important: after it, Chern coefficients are literal sums of chamber coefficients, with no residual global sign.

The rational function

$$K_d(z) := \mathcal{E}_C \left(\frac{\Delta_d(z) \mathcal{Q}_d(z)}{\mathcal{D}_d(z)} \right) = \sum_{\substack{a \in \mathbb{Z}^d \\ a_1 + \cdots + a_d = 0}} k_d(a) z_1^{a_1} \cdots z_d^{a_d} \quad (2)$$

is homogeneous of degree zero. The original strong Laurent-positivity statement is $k_d(a) \geq 0$ for all a . It is true through $d = 4$ and false at $d = 5$.

3 Chern packets, the universal kernel, and folding

3.1 The exact packet formula

For a partition $\lambda = (\lambda_1, \dots, \lambda_s) \vdash dL$ with $s \leq d$, pad λ by zeros to length d . Let $\mathrm{Orb}_{\mathrm{dist}}(\lambda)$ denote the set of *distinct* permutations of this padded d -tuple.

Proposition 3.1 (Chern-packet formula). *For every $d, L \geq 1$,*

$$[c_\lambda] \mathrm{Tp}_{A_d}^{L-1} = \sum_{\mu \in \mathrm{Orb}_{\mathrm{dist}}(\lambda)} k_d(\mu - L\mathbf{1}). \quad (3)$$

If $\ell(\lambda) > d$, the coefficient is zero.

Proof. A term z^a in (2), multiplied by $c_{j_i} z_i^{-j_i} z_i^{L-1}$, contributes a z_i^{-1} -term precisely when

$$j_i = L + a_i.$$

Since $\sum_i a_i = 0$, the selected indices sum to dL . For a prescribed commutative monomial c_λ , the labelled variables z_i realize exactly the distinct assignments of the padded index multiset λ . If an index is repeated, no stabilizer factorial is introduced. The residue signs were already cancelled above. This proves (3). $\square$

Remark 3.2. Formula (3) is the fundamental distinction between weak Chern positivity and strong Laurent positivity. Negative individual $k_d(a)$ are allowed, provided that every complete orbit packet is nonnegative.

3.2 The universal series kernel

Let

$$\Lambda_0 = \{a \in \mathbb{Z}^d : \sum_i a_i = 0\}$$

and let $L_0 = \mathbb{Z}^{\Lambda_0}$ denote coefficient arrays for which all coefficients under discussion are defined. Define the orbit-forgetting map

$$\varepsilon_+ : L_0 \longrightarrow \mathbb{Z}^{\Lambda_0/S_d}, \quad (\varepsilon_+ f)([a]) = \sum_{b \in \text{Orb}_{\text{dist}}(a)} f(b).$$

Proposition 3.3 (Universal kernel). *Let $P(z)$ be a homogeneous polynomial of the same degree as $\mathcal{Q}_d$, and put*

$$K_P = \mathcal{E}_C \left(\frac{\Delta_d P}{\mathcal{Q}_d} \right).$$

Replacing $\mathcal{Q}_d$ by $\mathcal{Q}_d + P$ changes no Thom polynomial at any level if and only if

$$\varepsilon_+(K_P) = 0.$$

At the coefficient-array level,

$$\ker \varepsilon_+ = \sum_{i=1}^{d-1} (1 - s_i) L_0, \tag{4}$$

where s_i exchanges coordinates $i, i+1$.

Proof. The forward implication follows from (3). Conversely, given any zero-sum charge orbit $[a]$, choose

$$L \geq \max(1, -\min_i a_i).$$

Then all $L + a_i$ are valid Chern indices, so the corresponding Chern coefficient tests precisely the orbit sum of a . Thus all levels detect every orbit. Finally, on each finite S_d -orbit, the kernel of the sum map is generated integrally by differences of vertices; adjacent transpositions generate all such differences, which gives (4). $\square$

Equation (4) is a complete abstract description of the universal kernel after chamber expansion. It does *not* by itself construct a polynomial numerator P whose rational chamber expansion realizes a desired difference array. Chamber-safe rational representatives are the difficult part.

3.3 0-Hecke folding

Define an adjacent sorting operator Π_i on coefficient arrays: off the wall $a_i = a_{i+1}$, it adds the two coefficients in the $\{1, s_i\}$ -orbit and places their sum at the dominant ordering; on the wall it retains the coefficient once. Then

$$\Pi_i^2 = \Pi_i, \quad \Pi_i \Pi_{i+1} \Pi_i = \Pi_{i+1} \Pi_i \Pi_{i+1},$$

so Π_i give the 0-Hecke action. If w_0 is the longest permutation, then

$$\Pi_{w_0} K_d \tag{5}$$

has, at each dominant charge, exactly the distinct orbit sum occurring in (3). Hence Rimányi positivity is equivalent to coefficientwise nonnegativity of $\Pi_{w_0} K_d$.

This is chamber-safe: one first expands K_d in the prescribed chamber and only then folds its coefficient array. It does not permute the rational function and re-expand it in a different chamber.

3.4 The symmetric-function transform

Let X be an infinite alphabet and

$$H(X/u) = \prod_{p \geq 1} (1 - x_p/u)^{-1} = \sum_{m \geq 0} h_m(X) u^{-m}, \quad h_m = 0 \ (m < 0).$$

Define

$$\mathfrak{H}_{d,L}(X) = \text{CT}_C \left[(z_1 \cdots z_d)^L K_d(z) \prod_{j=1}^d H(X/z_j) \right]. \tag{6}$$

Then

$$\mathfrak{H}_{d,L}(X) = \sum_{a \in \Lambda_0} k_d(a) \prod_{i=1}^d h_{L+a_i}(X). \tag{7}$$

Since $\text{Sym} = \mathbb{Z}[h_1, h_2, \dots]$, the substitution

$$h_j \mapsto c_j$$

sends $\mathfrak{H}_{d,L}$ exactly to $\text{Tp}_{A_d}^{L-1}$. Thus the conjecture is also exactly the h -monomial positivity of $\mathfrak{H}_{d,L}$, not Schur positivity.

4 Complete fixed-rank results through $d = 5$

4.1 Ranks $d \leq 3$

The cases $d = 1, 2$ are elementary. For all $L \geq 1$,

$$\text{Tp}_{A_2}^{L-1} = c_L^2 + \sum_{j=1}^L 2^{j-1} c_{L-j} c_{L+j}. \tag{8}$$

For $d = 3$, in simple-root chamber variables x, y , the normalized series is

$$H_3(x, y) = \frac{(1-x)(1-y)(1-xy)}{(1-2x)(1-2xy)(1-y-xy)}. \tag{9}$$

Its coefficients have an explicit positive negative-binomial-tail formula. In particular $H_3 \in \mathbb{Z}_{\geq 0}[[x, y]]$, proving all-level positivity for $d = 3$.

4.2 Rank $d = 4$

The saturated orbit closure is the associativity hypersurface and

$$\mathcal{Q}_4 = 2z_1 + z_2 - z_4.$$

Putting

$$x = z_1/z_2, \quad y = z_2/z_3, \quad z = z_3/z_4,$$

the normalized chamber series is

$$\begin{aligned} R_4(x, y, z) = & \frac{(1-x)(1-xy)(1-xyz)(1-y)(1-yz)(1-z)}{(1-2x)(1-2xy)(1-y-xy)(1-2xyz)} \\ & \times \frac{1-yz-2xyz}{(1-yz-xyz)(1-2yz)(1-z-xyz)}. \end{aligned} \quad (10)$$

The all-degree geometric-series/Stanley decomposition of (10) is coefficientwise nonnegative. This recovers the strong Laurent statement in rank four and therefore proves Chern positivity for every L . Importantly, individual coordinate-prime pieces of $\mathcal{Q}_4$ need not be Chern-positive; the full associativity circuit must already be grouped.

4.3 Rank $d = 5$: failure of the original strong statement

For the original numerator,

$$\left[\frac{z_1 z_3}{z_4 z_5} \right] \mathcal{E}_C \left(\frac{\Delta_5 \mathcal{Q}_5}{\mathcal{D}_5} \right) = -1.$$

Thus strong Laurent positivity is false. Nevertheless the complete Chern packet containing this term is positive.

Use the weights

$$\begin{aligned} B &= 2z_1 - z_2, & F &= 2z_1 - z_3, & E &= z_1 + z_2 - z_3, \\ D &= 2z_2 - z_4, & M &= z_2 + z_3 - z_5, & C &= z_1 + z_4 - z_5, \\ G &= 2z_1 + z_2 - z_5, & J &= 2z_1 + z_3 - z_5. \end{aligned}$$

The exact saturated multidegree factorization is

$$\mathcal{Q}_5 = G(DJ + BC) = GDJ + GBC. \quad (11)$$

4.4 The universal GBC cancellation

Let

$$S = \{2z_1, z_1 + z_2, 2z_2, z_1 + z_3\}, \quad P(t) = \prod_{\alpha \in S} (\alpha - t).$$

Then the complete thirteen-factor denominator is

$$\mathcal{D}_5 = BFE P(z_4) P(z_5) M(z_5) C. \quad (12)$$

Also $G = F + M$. For a finite truncation

$$\Phi_{\ell, N}(z) = \prod_{i=1}^5 z_i^\ell \left(\sum_{j=0}^N c_j z_i^{-j} \right),$$

consider the residue against the symmetric insertion $\Phi_{\ell, N}$.

First,

$$\frac{BMC}{\mathcal{D}_5} = \frac{1}{FEP(z_4)P(z_5)}$$

is invariant under $z_4 \leftrightarrow z_5$. All its z_4, z_5 poles lie on the inner z_1, z_2, z_3 scale. Hence the two outer contours can be interchanged without crossing a pole. The Vandermonde changes sign, while the remaining integrand and the product measure do not; the residue is zero.

Second,

$$H := \frac{FBC}{\mathcal{D}_5} = \frac{1}{EP(z_4)P(z_5)M(z_5)}.$$

Its s_{45} -symmetric part vanishes by the same outer-contour swap. For the antisymmetric part, set

$$T(t) = P(t)M(t).$$

Direct subtraction, before any chamber expansion, gives

$$H - s_{45}H = \frac{z_5 - z_4}{ET(z_4)T(z_5)}. \quad (13)$$

The completed source multiset

$$\{2z_1, z_1 + z_2, 2z_2, z_1 + z_3, z_2 + z_3\}$$

and E are invariant under $z_1 \leftrightarrow z_2$. Every nonzero z_1, z_2 pole of (13) lies on an outer z_3, z_4, z_5 scale. The two inner contours may therefore be interchanged safely, and (13) also has zero residue. Since

$$GBC = (F + M)BC = FBC + BMC,$$

we obtain

$$\text{IRes}_C \frac{\Delta_5 GBC}{\mathcal{D}_5} \Phi_{\ell, N}(z) dz_1 \cdots dz_5 = 0 \quad (14)$$

for every ℓ, N . Passing to arbitrarily large N proves universal kernel membership for the full Chern series.

This proof was independently reconstructed with explicit nested radii, for example $R_{i+1} > 10R_i$. It handles every level, every charge, repeated indices, zeros, and the convention $c_j = 0$ for $j < 0$. It is not a permutation of an already-expanded Laurent series.

4.5 Strong positivity of the modified representative GDJ

Put

$$x = z_1/z_2, \quad b = z_2/z_3, \quad c = z_3/z_4, \quad \delta = z_4/z_5, \quad r = c\delta, \quad q = br, \quad p = xbr.$$

After cancelling D , the normalized chamber series for GDJ factors as

$$W\mathcal{C},$$

where

$$\begin{aligned} W &= \frac{1-b}{1-b-xb} \frac{1-bc}{1-bc-xbc} \frac{1-c}{1-c-xbc} \\ &\times \frac{1-\delta}{1-\delta-p} \frac{1-p}{1-p-q} \frac{1-xbc}{1-2xbc} \end{aligned} \quad (15)$$

with multiplication understood between all displayed factors, and

$$\mathcal{C} = \frac{N}{ASPQUV},$$

where

$$\begin{aligned} A &= 1 - 2x, & S &= 1 - 2xb, & P &= 1 - 2p, & Q &= 1 - 2q, \\ U &= 1 - r - p, & V &= 1 - r - q. \end{aligned}$$

A direct common-denominator identity gives the Stanley decomposition

$$\begin{aligned} \mathcal{C} &= 1 + \frac{x}{A} + \frac{xb}{S} + \frac{x^2b}{AS} + \frac{br + br^2}{Q} + \frac{xb^2r}{SQ} \\ &\quad + \frac{x^2br}{AU} + \frac{x^3b^2r}{ASU} + \frac{x^2b^3r^2}{SQU} + \frac{xb^2r^3}{PV} + \frac{br^3}{QV}. \end{aligned} \tag{16}$$

Every ratio in W is of the form

$$\frac{1-u}{1-u-v} = 1 + \frac{v}{1-u-v} \quad \text{or} \quad \frac{1-u}{1-2u} = 1 + \frac{u}{1-2u},$$

and every inverse in (16) is the geometric series of a polynomial with nonnegative coefficients. Hence

$$\mathcal{E}_C\left(\frac{\Delta_5 GDJ}{\mathcal{D}_5}\right) \in \mathbb{Z}_{\geq 0}[[x, b, c, \delta]].$$

Together with (14), this proves Chern positivity for $d = 5$ and all L .

Remark 4.1 (A stabilizer-safe calibration). At $L = 1$, the packet $\lambda = (2, 2, 1, 0, 0)$ has $5!/(2!2!) = 30$ assignments, not 120. Only five ordered residues are nonzero:

ordered indices	22100	12200	21200	22010	22001
coefficient	6	1	-1	2	2.

Their sum is 10. The GBC correction, when grouped by the last charge, is $(-2, 2, 0)$: it is globally null but not null with the last factor marked. This is the first concrete sign that the successful kernel mechanism is nonlocal.

5 The curvilinear-incidence formulation

5.1 The incidence formula

Let $n = d + 1$, and let

$$C_{n,r} = \overline{\text{Hilb}_0^{n,\text{curv}}(\mathbb{A}^r)}$$

be the projective curvilinear component of the punctual Hilbert scheme. Let $\mathcal{V}$ be the rank- n tautological algebra and

$$\mathcal{E} = \ker(\mathcal{V} \rightarrow \mathcal{O})$$

its rank- d augmentation bundle. Put $q = r + \ell$.

Theorem 5.1 (Incidence formula). *For target Chern roots $x_1, \dots, x_q$,*

$$\text{Tp}_{A_d}^\ell(e_1(x), e_2(x), \dots) = \int_{C_{n,r}} \prod_{a=1}^q e_T(\mathcal{E} \otimes \mathbb{C}_{x_a}). \tag{17}$$

Proof sketch. Choose a finite jet space $W_{r,N}$ with $N \geq n$. Universal evaluation gives a surjection

$$W_{r,N} \otimes \mathcal{O}_{C_{n,r}} \twoheadrightarrow \mathcal{E}.$$

The incidence of q -tuples of functions vanishing on the universal subscheme is the total space of the kernel of the corresponding surjection onto $\mathcal{E} \otimes \mathbb{C}^q$. It is irreducible, and its projection has codimension

$$qd - \dim C_{n,r} = d(q - r + 1) = d(\ell + 1).$$

Over the open curvilinear locus, a general q -tuple generates the ideal and determines its length- n zero scheme uniquely. The projection is therefore generically one-to-one onto the Morin stratum. Its equivariant Euler class is the integrand in (17). $\square$

5.2 Extraction of an individual Chern coefficient

Let $\lambda = (\lambda_1, \dots, \lambda_s) \vdash dL$, $s \leq d$, and assume $q \geq \lambda_1$. Define

$$\nu_{r,L}(\lambda) = \text{sort}(q^{d-s}, q - \lambda_1, \dots, q - \lambda_s). \quad (18)$$

Proposition 5.2 (Characteristic-number formula).

$$[c_\lambda] \text{Tp}_{A_d}^{L-1} = \int_{C_{n,r}} m_{\nu_{r,L}(\lambda)}(\mathcal{E}). \quad (19)$$

Indeed, if $\alpha_1, \dots, \alpha_d$ are formal Chern roots of $\mathcal{E}$, then

$$\prod_{i=1}^d \prod_{a=1}^q (x_a + \alpha_i) = \prod_{i=1}^d \left(\sum_{j=0}^q e_j(x) \alpha_i^{q-j} \right).$$

The coefficient of $e_{\lambda_1} \cdots e_{\lambda_s}$ is exactly the monomial symmetric function in (19). Repeated exponents are summed over distinct assignments, so again there is no stabilizer factorial.

5.3 The multiplicity-one hyperplane identity

The coordinate u_r induces a tautological section of $\mathcal{E}$. Its zero scheme consists of subschemes contained in the hyperplane $\mathbb{A}^{r-1} \subset \mathbb{A}^r$.

Lemma 5.3 (Hyperplane lemma).

$$c_d(\mathcal{E}) \cap [C_{n,r}] = [C_{n,r-1}] \quad (20)$$

in the Chow group of $C_{n,r}$.

Proof. The support assertion follows by projecting a DVR degeneration of curvilinear schemes to the hyperplane and applying Nakayama's lemma. On the standard curvilinear chart

$$I = (u_2 - p_2(t), \dots, u_r - p_r(t), t^n), \quad p_r(t) = \sum_{k=1}^d a_{rk} t^k,$$

the section is the regular d -tuple $(a_{r1}, \dots, a_{rd})$. Its generic zero is transverse, so the unique component occurs with multiplicity one. $\square$

If $\lambda_1 \leq q - 1$, all entries of $\nu_{r,L}(\lambda)$ are positive and

$$m_{\nu_{r,L}(\lambda)}(\alpha) = (\alpha_1 \cdots \alpha_d) m_{\nu_{r-1,L}(\lambda)}(\alpha).$$

Equations (19) and (20) therefore descend the coefficient from source dimension r to $r - 1$.

5.4 The all-rank positive sector

Theorem 5.4 (Uniform sector positivity). *For every $d, L \geq 1$,*

$$\lambda_1 \leq L + 1 \implies [c_\lambda] \text{Tp}_{A_d}^{L-1} \geq 0.$$

Proof. Repeatedly apply the hyperplane lemma until $r = 2$. Then $|\nu| = d$ and

$$[c_\lambda] \text{Tp}_{A_d}^{L-1} = \int_{C_{n,2}} m_\nu(\mathcal{E}).$$

Briançon's theorem identifies $C_{n,2} = \text{Hilb}_0^n(\mathbb{A}^2)$. Under the Lehn–Sorger associated-graded identification, the Chern classes of $\mathcal{E}$ correspond to the elementary symmetric functions of the Jucys–Murphy elements

$$J_k = \sum_{a < k} (a \ k) \in \mathbb{Z}[S_n].$$

For $|\nu| = d = n - 1$, the central element $m_\nu(J_1, \dots, J_n)$ has nonnegative coefficients in the permutation basis. Its coefficient at a fixed n -cycle counts minimal transposition words with exponent signature ν . Top degree on the punctual Hilbert scheme is one-dimensional, and $\int c_d(\mathcal{E}) = 1$. Hence the desired integral is precisely this nonnegative count. $\square$

As a calibration, on the plane one obtains the Narayana formula

$$\int_{C_{n,2}} c_i(\mathcal{E}) c_{d-i}(\mathcal{E}) = \frac{1}{d+1} \binom{d+1}{i} \binom{d+1}{i+1}.$$

Equivalently,

$$[c_2^k c_1^{d-2k}] \text{Tp}_{A_d}^0 = \binom{d}{2k} \frac{1}{k+1} \binom{2k}{k}.$$

6 Nested incidence, exact recursions, and their obstruction

Let

$$F_{n-1,n;r} \subset C_{n-1,r} \times C_{n,r}$$

be the reduced closure of the graph assigning to a curvilinear length- n scheme its unique length- $(n-1)$ truncation, with projections α, β . There is an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \alpha^* \mathcal{E}_n \longrightarrow \beta^* \mathcal{E}_{n-1} \longrightarrow 0, \quad \zeta = c_1(\mathcal{L}).$$

Define the main Segre cycles

$$S_k^{n,r} = \beta_* (\zeta^{r-1+k} [F_{n-1,n;r}]).$$

If μ is a partition of $(n-1)(r-1)$, then splitting off the distinguished root gives the exact recursion

$$\int_{C_{n,r}} m_\mu(\mathcal{E}_n) = \sum_{\substack{a \in \text{Dist}(\mu) \\ a \geq r-1}} \int_{C_{n-1,r}} m_{\mu \setminus a}(\mathcal{E}_{n-1}) S_{a-r+1}^{n,r}. \quad (21)$$

The sum is over distinct values of positive parts; repeated parts do not create multiplicities.

Keeping the complete target-root factor gives another exact identity:

$$\beta_* \prod_{a=1}^q (\zeta + x_a) = \sum_{j=0}^L c_j S_{L-j}^{n,r}.$$

Consequently

$$\text{Tp}_{A_d}^{L-1} = c_L \text{Tp}_{A_{d-1}}^{L-1} + \sum_{k=1}^L c_{L-k} \mathfrak{D}_{d-1,k}^{L,r}, \quad (22)$$

for explicitly defined Segre descendants $\mathfrak{D}_{d-1,k}^{L,r}$.

Equations (21)–(22) are genuine all-rank recursions, but they do not close in a known positive state basis. Three exact obstructions were found.

1. After a descendant has positive codimension, the next transition has a larger index range. A zero-exponent branch is genuinely nonzero; in the first instance it equals 2^{r-2} .

2. The first new class $S_2^{5,3}$ has no rational representative as a symmetric degree-two polynomial in the Chern roots of the lower augmentation bundle. Such a representative $A m_{(2)} + B m_{(1,1)}$ would have to satisfy

$$12A + 6B = 11, \quad 12A + 12B = 10, \quad 10A + 5B = 9,$$

an inconsistent system.

3. On the effective square-zero boundary $B \simeq \text{Gr}(2, 3) \simeq \mathbb{P}^2$, the augmentation bundle is the universal quotient $\mathcal{Q}$, and

$$m_{(2,0)}(\mathcal{Q}) = c_1(\mathcal{Q})^2 - 2c_2(\mathcal{Q}) = -h^2.$$

Thus individual effective boundary states can have negative Chern-monomial evaluation.

A delayed-history calculation does produce nonnegative transition weights through the first associativity diamond, uniformly in the source dimension, and recovers $d = 4$. The state remembers an unresolved transition index until a later projective push. This is evidence for a history-sensitive algebra, but no closed all-length version was found.

7 Circuits, balancing, and the first-fold criterion

7.1 Circuit exchange

Suppose a homogeneous multidegree circuit has a positive identity

$$\sum_A u_A \prod_{q \in A} \beta_q = \sum_B v_B \prod_{q \in B} \beta_q, \quad u_A, v_B \in \mathbb{Z}_{\geq 0}.$$

After division by the common ambient product $\prod_q \beta_q$, fixed-chamber expansion gives an exact equality of positive denominator-word operators. Multiplication by the common Vandermonde and application of the Frobenius map in (7) preserve this equality. This construction is chamber-safe, but it gives equality rather than positivity of the individual circuit pieces.

At $d = 4$, with

$$\begin{aligned} A &= 2z_1 - z_2, & D &= 2z_2 - z_4, \\ B &= z_1 + z_2 - z_3, & C &= z_1 + z_3 - z_4, \end{aligned}$$

the first circuit is

$$A + D = B + C = \mathcal{Q}_4.$$

One genuine coordinate-prime component already satisfies

$$[h_{3L} h_L] = -2 \cdot 6^{L-1} < 0.$$

Thus the entire first circuit must be clustered.

At $d = 5$, the Hilbert–Burch circuit is

$$BE + BM + DM = DJ + BC.$$

The additional fact that GBC vanishes after full Frobenius multiplication is not a consequence of this polynomial identity alone; it is the global two-swap relation proved above.

7.2 The balancing ideal

For a split $d = p + q$, retain the Chern factors in two alphabets:

$$\mathfrak{f}_{p|q}[\alpha] = \prod_{i \leq p} h_{\alpha_i}(X) \otimes \prod_{i > p} h_{\alpha_i}(Y).$$

The ordinary Frobenius map is multiplication after identifying alphabets:

$$\mathfrak{f}_d = \mu \circ \mathfrak{f}_{p|q}.$$

Since $\text{Sym} = \mathbb{Z}[h_1, h_2, \dots]$,

$$\ker \mu = \langle h_r(X) \otimes 1 - 1 \otimes h_r(Y) : r \geq 1 \rangle. \quad (23)$$

The $d = 5$ correction is already nonlocal in this exact sense. At $L = 1$, in the charge multiset $(-1, -1, 0, 1, 1)$, its pointed S_4 -packet values according to the last charge are

$$(-2, 2, 0).$$

Thus

$$\mathfrak{f}_{4|1}(GBC) = 2h_2^2(1 \otimes h_1 - h_1 \otimes 1) \neq 0,$$

while full multiplication kills it by (23). Any induction that requires the correction to vanish before the terminal Chern factor is unmarked is therefore false already in rank five.

7.3 A precise first-fold target

There is nevertheless a useful exact rank-one criterion for the first 0-Hecke fold. Encode a zero-sum charge by prefix coordinates

$$n_j = a_1 + \dots + a_j.$$

Fix $n_2 = q$, put $x = z_1/z_2$, and write one coefficient string as

$$F = (1 - x)H, \quad H = \sum_m h_m x^m.$$

For $m \geq q/2$, set

$$S_m = h_m - h_{q-m-1}.$$

Then the first folded coefficient is

$$[x^m]\Pi_1 F = \begin{cases} S_m, & q = 2m, \\ S_m - S_{m-1}, & q < 2m. \end{cases}$$

Consequently the first fold is nonnegative exactly when the reflected cumulative quantities S_m start nonnegative at the wall and are nondecreasing away from it. This is strictly weaker than raw Laurent positivity and gives a concrete target for a path or minimal-parabolic model. No all- d proof of this monotonicity was found.

8 Other rigorous partial structures

8.1 Stable Thom-series and cycle-sector information

The all-level family is injectively encoded by its stable Thom series: positivity at every level is equivalent to positivity of every stable coefficient. Several coarser specializations have exact positive combinatorics. In particular, the total coefficient mass and the distribution by the number of Chern factors agree with permutation-cycle statistics:

$$\sum_{\ell(\lambda)=r} [c_\lambda] \text{TP}_{A_d}^{L-1} = (d!)^{L-1} \begin{bmatrix} d \\ r \end{bmatrix},$$

where $\begin{bmatrix} d \\ r \end{bmatrix}$ is an unsigned Stirling number of the first kind. Thus the total mass is $(d!)^L$.

These identities do not determine the distribution among Chern partitions of a fixed length. The first mismatch with the naive cycle-type polynomial occurs at $d = 4, L = 1$, and for $L > 1$ already at $d = 2$. Noncrossing/free-cumulant reorganizations give additional positive scalar and cycle-sector recurrences, but a shape-refined positive gluing rule was not obtained.

8.2 The nonassociative tower point class

On the source-rank-one nonassociative tower, the orbit point has the exact class

$$P_d^\star = \prod_{k=4}^d \prod_{\substack{2 \leq i \leq j \\ i+j \leq k}} (z_1 + z_{i-1} + z_j - z_k),$$

while the point cycle itself is $(-1)^{N_d} P_d^\star$, where N_d is the number of displayed factors. The polynomial $P_d^\star$ recovers $\mathcal{Q}_4$, and at $d = 5$ its Chow class equals that of GDJ on the rank-one tower.

This identity does not stabilize to the universal residue problem. At $d = 5$,

$$P_5^\star - \mathcal{Q}_5 = GBE,$$

and this difference has the negative level-two coefficient $[c_7 c_3] = -4$; it is not in the universal kernel. A related factored $d = 6$ candidate agrees in one tested level but gives $[c_3^4] = -2$ at the next. These examples are useful warnings against promoting rank-one Chow relations or finite-level agreement to a universal identity.

9 Registry of blocked proof families

The following table records mathematically distinct routes and the first decisive obstruction found. “Blocked” means that the stated mechanism, in its tested precise form, cannot prove the conjecture; it does not exclude every possible refinement.

Route	Exact obstruction
Original strong Laurent positivity	False at $d = 5$: $[z_1 z_3 / (z_4 z_5)] K_5 = -1$.
Local adjacent contour-swap kernel	The universal coefficient-array kernel is generated by adjacent differences, but rationally safe adjacent swaps do not generate the GBC correction. The successful $d = 5$ proof needs two nonlocal swaps and completion of a source multiset.

Route	Exact obstruction
Coordinate-prime or facetwise positivity	A genuine $d = 4$ component has $[h_{3L}h_L] = -2 \cdot 6^{L-1}$; two $d = 5$ radical components have negative full packets. Positivity exists only after circuit grouping.
Unsaturated associativity/SAGBI	The quadratic associativity scheme acquires extra components from $d = 6$. Its coordinate-prime multidegrees do not equal the saturated Morin orbit multidegree.
Ordinary geometric effectivity or Schur positivity	Chern extraction involves signed inverse-Kostka transformations. On the effective square-zero boundary, $m_{(2)}(\mathcal{Q}) = -h^2$.
Nef-root or mixed-volume argument	The augmentation bundle is globally generated, but it admits no birational splitting into nef root lines on the first square-zero boundary. Even Lorentzian mixed-degree arrays can give $\int m_{(2,0)} = -1$.
One-index nested-incidence induction	Descendants do not close, the zero branch is nonzero, and $S_2^{5,3}$ has no tautological symmetric representative.
Ordinary Jucys/partial-permutation state	At $(n, r) = (4, 3)$, two exact linear relations among partial-Jucys elements evaluate to 1, not 0, on the geometric characteristic numbers. Final support and final permutation forget necessary history.
Shape-local Callan or cumulant gluing	At $d = 6$, integer length-preserving bilinear gluing forces even length-two coefficients, while $R_6^{(2)} = 58c_{51} + 19c_{42} + 5c_{33}$ contains odd coefficients.
Restriction interpolation/total positivity	A natural $d = 6$ restriction block has forced minors -10 and -88 under every admissible ordering and sign reorientation.
Generic invariant-cone separation	The stable $I_{2,2}$ cone contains the negative Chern direction $c_3^2 - c_2c_4$. Generic cone effectivity cannot characterize the Morin Chern cone.
Toric resonance, NRGIT leaves, or boundary subtrees	Individual $d = 4$ circuit cones and $d = 5$ prism/leaf strata have negative packets. The unipotent cancellation loci are not separated by the diagonal resonance fan.
Multiple-point effectivity	Extracting the primitive Morin class uses partition-lattice logarithmic inversion; already $\Sigma_2 = n_2 - n_1^2$, so effectivity of the full multiple-point class does not give coefficientwise positivity.
Hall, LLT, or standard commutative CoHA models	Exact coefficient mismatches occur at $(d, L) = (2, 2)$ or $(3, 2)$. Complete-graph LLT cumulants retain an unavoidable e_{21} term when the Morin primitive is $2e_3$.

10 Most plausible continuation mechanisms

At the stopping point, four genuinely different mechanisms remained mathematically viable.

Global coefficient folding. Prove directly that $\Pi_{w_0} K_d \geq 0$, perhaps by a path model for the reflected cumulative strings. This route targets exactly the Chern coefficients and automatically handles stabilizers. It needs a global argument; positivity of the unfurled denominator series is insufficient.

Circuit straightening modulo balancing. Use saturated multidegree circuits to rewrite the positive denominator-word sum, then show that every signed remainder is congruent modulo the full balancing ideal (23) to an h -positive expression. The $d = 5$ *GBC* cancellation is the first model. Any block-local version is disproved by the pointed $(-2, 2, 0)$ split.

History-sensitive incidence algebra. Construct a positive state space that remembers unresolved collision indices long enough for generic and rank-jump boundary contributions to cancel. It must be coarser than the basis of individual effective strata, but finer than final-support partial permutations.

Saturated cyclic/derived associativity. Describe $\mathcal{Q}_d$, not the unsaturated associativity scheme, by a determinantal, matroidal, or cellular object whose complete circuit clusters are compatible with chamber extraction. Low-rank identities $\mathcal{Q}_4 = A + D$ and $\mathcal{Q}_5/G = BE + BM + DM = DJ + BC$ strongly suggest such a structure, but no all- d saturated formula was found.

11 Audit of principal failure points

The following checks were applied to every positive claim retained in this report.

1. **Residue signs.** With $\text{Res}_\infty = -[z^{-1}]$, the d residue signs cancel the outer $(-1)^d$. Zero identities such as the *GBC* cancellation are valid before this common sign is applied.
2. **Chamber order.** No proof permutes a rational function and silently re-expands it in the original chamber. The $d = 5$ swaps are contour homotopies with explicitly separated poles; the folding operator acts only after fixed-chamber expansion.
3. **Repeated indices.** Every packet and monomial symmetric function uses distinct assignments. Repeated entries reduce the orbit size; they never introduce a stabilizer factorial.
4. **Zero and negative indices.** The convention $c_0 = 1$ is included in all truncations and packet formulas. Terms requesting c_j with $j < 0$ vanish.
5. **All levels.** Universal kernel membership is tested on all zero-sum charge orbits, equivalently on all L after a sufficiently large shift. The *GBC* proof works against an arbitrary symmetric truncation and hence for every ℓ , not just computed values.
6. **No finite-pattern inference.** Exact calculations were used to find and reject identities. The retained $d = 5$ positivity rests on the polynomial identity (16); the sector theorem rests on the geometric hyperplane and Jucys arguments. A $d = 6$ candidate that passed one level and failed the next was discarded.
7. **Basis distinctions.** Schur positivity, root-monomial positivity, effectivity, nefness, and Laurent positivity are not substituted for Chern-monomial positivity. Explicit signed change-of-basis and boundary examples separate these notions.
8. **No circularity.** The remaining candidate mechanisms are stated as open mechanisms, not lemmas assumed from low-rank evidence. In particular, no unproved positive-representative theorem for general $\mathcal{Q}_d$ is used.

12 Final assessment

The proof search materially sharpened the problem but did not resolve the all-rank conjecture. Its strongest complete achievements are:

- full positivity through $d = 5$, including a rigorous repair of the first failure of Laurent positivity;
- a general geometric coefficient formula and a nontrivial all- d, ℓ positive sector;
- exact algebraic descriptions of the Chern packets, the universal series kernel, and the global balancing quotient;
- decisive counterexamples to componentwise, local, and generic effectivity arguments.

The central unresolved phenomenon is now precise. Starting at rank five, negative ordered chamber contributions are repaired by relations that move Chern weight between labelled factors and vanish only after full commutative symmetrization. Starting at rank six, the raw associativity scheme also ceases to equal the saturated Morin orbit. A complete proof must control both effects simultaneously: saturation of the orbit multidegree and nonlocal balancing of complete Chern packets.

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